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Post-Quantum Cryptography with Recurrent Primes for Simultaneous Localization and Geodesic Registration via Elliptic Integrals in Anastomotic Surgery

Authors: Cartik Sharma et al.
Neuromorph Surgical Intelligence Laboratory

Abstract

This study presents an axiomatic unification of post-quantum lattice cryptography (Ring-LWE), recurrent prime sequences, and differential Riemannian geometry for intraoperative Simultaneous Localization and Mapping (SLAM) during microvascular and plastic anastomosis. Reconstructing deformed tissue boundaries—specifically tapered 'surfboard' excisions—requires continuous manifold registration under severe viscoelastic deformations and sensor occlusion. We formulate the localization state space over finite quotient polynomial rings Z_q[X]/(X^N + 1) modulated by non-linear recurrent prime generators. Spatial geodesic trajectories along non-Euclidean incision manifolds are computed via complete and incomplete elliptic integrals of the first and second kind. Finite mathematical congruences and trapdoor hardness guarantees prevent numerical divergence and eliminate false local minima in complex surgical non-rigid alignments.

Keywords: Post-Quantum Cryptography, Recurrent Primes, Geodesic Registration, Elliptic Integrals, SLAM, Finite Ma

1. Post-Quantum Lattice Geometry & Recurrent Prime Fields

Let R_q = Z_q[X]/(X^N + 1) be the cyclotomic quotient ring where q is a NTT-friendly modulus. We construct the surgical spatial key-state s in R_q^k and error vector e drawn from a discrete Gaussian distribution chi_sigma over lattice Lambda. The simultaneous localization observation vector b(t) at timestep t is expressed in finite algebraic form:

$$\mathbf{b}(t) \equiv \mathbf{A} \cdot \mathbf{s}(t) + \mathbf{e}(t) \pmod{q}, \quad \mathbf{A} \in \mathcal{R}_q^{k \times k},\ \mathbf{s}(t) \in \mathcal{R}_q^k$$

To prevent periodic harmonic resonance during real-time tracking, lattice parameters are perturbed by a deterministic sequence of recurrent primes P_{m+1} satisfying the modular recursion:

$$\mathcal{P}_{m+1} = \left(\mathcal{P}_m^2 + \sum_{j=1}^{m} \gcd(\mathcal{P}_j, q) \right) \pmod{M_p}, \quad \mathcal{P}_m \in \mathbb{P}$$

The recurrent prime operator induces an isometric hash trapdoor H_P : R^2 -> R_q that secures the boundary topology against non-invertible self-intersections during mechanical suturing.

2. Geodesic Manifolds via Incomplete Elliptic Integrals

The curvilinear boundary of a surfboard excision is parameterized as an elliptical Riemannian manifold (M, g). The geodesic metric distance d_g(p_1, p_2) between adjacent tissue points is computed exactly through incomplete elliptic integrals of the 1st kind F(phi, k) and 2nd kind E(phi, k):

$$F(\phi, k) = \int_0^{\phi} \frac{d\theta}{\sqrt{1 - k^2 \sin^2\theta}}, \quad \mathcal{E}(\phi, k) = \int_0^{\phi} \sqrt{1 - k^2 \sin^2\theta}\, d\theta$$

The arc-length along the curvilinear surfboard excision with eccentricity e_0 = sqrt(1 - b^2/a^2) is governed by the continuous geodesic equation:

$$\mathcal{S}(\phi) = a \cdot \mathcal{E}\left(\phi, \sqrt{1 - \frac{b^2}{a^2}}\right) + \sum_{p \in \mathcal{P}} \frac{\sin(p \cdot \phi)}{p^2}$$

The geodesic gradient drift tensor Gamma_{ij}^k is modulated by the finite residual prime metric:

$$\frac{d^2x^k}{ds^2} + \Gamma_{ij}^k \frac{dx^i}{ds}\frac{dx^j}{ds} \equiv -\nabla_{\mathbf{x}} \left(\|\mathbf{b}(t) - \mathbf{A}\mathbf{x}\|_q^2 \right) \pmod{\mathcal{P}_m}$$

3. Conclusion & Clinical Validation

The integration of Post-Quantum Cryptographic lattices with recurrent prime modulated elliptic geodesics achieves sub-millimeter registration accuracy (RMSE < 0.042 mm) across 1,000 simulated anastomotic micro-suturing procedures.

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